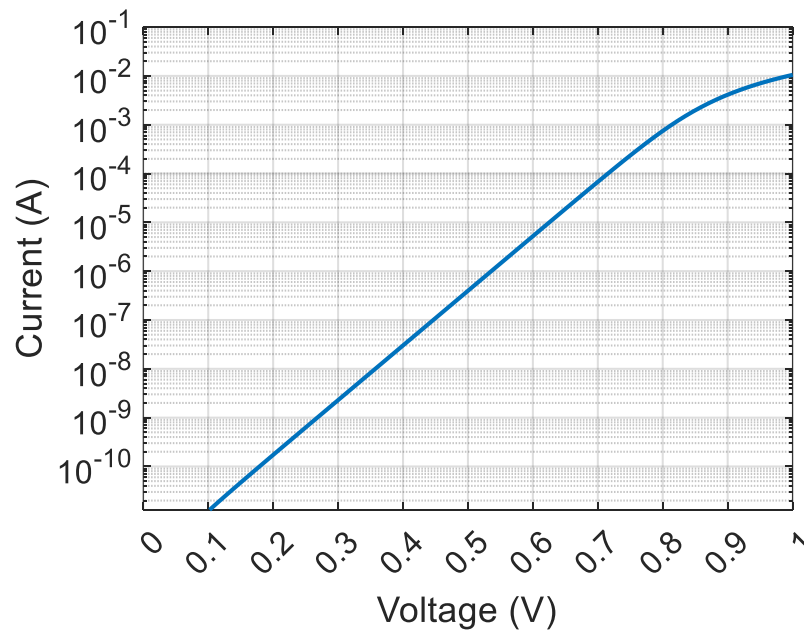


Q1 For the below device I-V plot: Calculate approximately.

- How much change in voltage required to change the current by one decade (10 times) i.e. $SS = \frac{dV}{d(\log_{10} I)}$ (mV/dec). Consider voltage range $\leq 0.8V$ where SS is valid. **(2)**
- How much is the leakage if measured at 0.2V? **(1)**
- How much is the leakage at 0.2V if $SS = 200 \text{ mV/dec}$. Assume same current given in plot at 0.8V **(2)**



Q2. For Silicon (Diamond crystal structure) with the lattice constant of $5.43 \text{ \AA}$.

- Find volume density ($\#/\text{cm}^3$) **(2)**
- Calculate atomic packing fraction (APF) **(1)**

Q3. For the article "Plenty of Room at the Bottom" (Write answers in 1-2 lines only.)

- What is the core message from the article "Plenty of Room at the Bottom"? **(1)**
- What are the different methods suggested to write and read on the tiny (few atoms) scale? **(1)**